

Finding a Feasible Point

Given : $P = \{x \in \mathbb{R}^n : a_i^T x \leq b_i, i=1, \dots, m\}$

Assume for now (can be removed)

P is bdd and if non-empty it is full dimensional.

Goal: In time polynomial in $\dim^n$, and the description length $\langle A \rangle, \langle b \rangle$, decide

A. Is P empty or not

B. If non-empty, point P .

The Basic Algorithmic Idea

- Suppose we have an oracle :

$x \rightarrow \square \rightarrow$ feasible or provides
(a_0, b_0) violated const.

- Suppose initially we know

a. $P \subseteq \underline{B(0, R^2)}$

$\rightarrow$ b. $\text{vol } P \geq \frac{2^{-(n+1)} \langle A \rangle + n^3}{A_0}$, if P non-empty

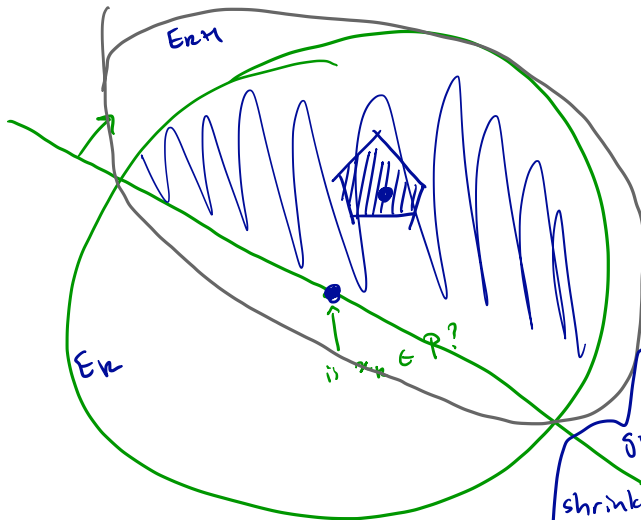
Initialize : $E_0 = \underline{B(0, R^2)} = \underline{E(\overset{A_0}{R^2 I}, \overset{\text{Center}}{\vec{0}})}$

At iteration k : $E_k = \underline{E(A_k, \overset{\text{Center}}{x_k})}$

Give x_k to oracle. If " $x_k \in P$ " — STOP.

ELSE : Find $\underline{E_{k+1} \supseteq E_k \cap \{x : \hat{a}^T x \leq \hat{b}\}} \supseteq \underline{P}$

By picture:



We need:

Way to find E_{n+1} to
guarantee its volume
shrinks quickly enough.

The Löwner-John Ellipsoid

Thm: For $K \subseteq \mathbb{R}^n$ convex, $\exists$ unique
minimum volume ellipsoid $E \supseteq K$.

Moreover, $K \supseteq \frac{1}{n} E$, where
" $\frac{1}{n} E$ " is the shrinkage of E w.r.t. center.

This is called the Löwner - John
ellipsoid of a set K .

For a general K convex, the LJ ellipsoid
may be hard to compute.

We will see that for $K = E \cap H_+$, L.J. ellipsoid of K
easy to compute.